JaeSeong Kim

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Research Interests

Computational imaging through joint optimization of optics, sensors, and algorithms. Differentiable ISP pipelines, polarization imaging, and end-to-end camera optimization for mobile, automotive, and microscopy applications.

Education

Tel Aviv University — Tel Aviv, Israel
M. Sc. in Electrical Engineering (94.19/100) — *April 2025*
Advisor: Prof. David Mendlovic

Tel Aviv University — Tel Aviv, Israel
B. Sc. in Electrical and Electronics Engineering (83.24/100) — *July 2022*

Research Experience

Tel Aviv University & Samsung Corephotonics (Joint industrial Ms.c)
Research Assistant (with Prof. David Mendlovic) — *Oct 2022-Aug 2025*
- Polarization imaging for mobile face anti-spoofing. Developed method achieving 10x lower error rate than RGB solutions. Simulated hybrid sensor architectures and recruited 64 participants for novel polarization face dataset.

Tel Aviv University
Research Assistant (with Prof. David Mendlovic) — *Jul 2021-Jul 2022*
Optics & Computer Vision Lab
- Developed CNN-based crime classification system using skeleton tracking to analyze human movements in surveillance videos, classifying actions into 10 crime categories.

Publication & Patents

"On the Effectiveness of Sparse Linear Polarization Pixels for Face Anti-Spoofing", **J. Kim**, A. Pelz, M. Scherer and D. Mendlovic, *IEEE Sensors Journal*, 2025

"Sensors, systems and methods for compact face identification polarization cameras", M. Scherer, **J. Kim**, Patent WO2024161329A1, 2024

"Enhanced Face Anti-Spoofing using Angle of Linear Polarization", **J. Kim**, M.Sc. Thesis, Tel Aviv University, 2025

Employment

Samsung Corephotonics — Tel Aviv, Israel
Image Quality Engineer — *Nov 2022-Apr 2025*
- Developed automated tool for quantifying EIS angular correction and motion blur, improving measurement accuracy by 25% and reducing analysis time from hours to 10 minutes
- Tuned ISP parameters using Qualcomm Chromatix and measured image quality with IMATEST

Pit-In Global Seoul, Korea
Software Developer (internship) *Jul 2017-Sep 2017*
- Created SQL database for location tracking by developing software for Raspberry Pi that leverages Bluez to collect Bluetooth signal strength indicators (RSSI) from customers' mobile devices

CoderBunker Shanghai, China
Community Manager *Jan 2016-Mar 2016*

Xinchejian(新车间) Shanghai, China
Community Manager *2015*

Presentations

Samsung Corephotonics Lecture Club
"Face Anti-spoofing using full polarization characteristics"
March 2025

Tel Aviv University Electrical Engineering Department Student Seminar
"Enhanced Face Anti-spoofing using Angle of Linear Polarization"
March 2025

Korean Embassy in Israel Research Seminar
"Camera Technologies - Multimodal cameras & Image Quality Assessment"
May 2024

Skills

- Technical Skills (Image Processing, Computer Vision, Deep Learning)
- Programming (Python, MATLAB, C/C++)
- Tools / Software (IMATEST, OpenCV, PyTorch, Qualcomm Chromatix)

Certifications

- Oracle Certified Professional, JAVA SE 6 Programmer
- Deep Learning Specialization (by Andrew Ng, deeplearning.ai)

Additional Information

- Military Service: KATUSA (Korean Augmentation to the United States Army), Republic of Korea Army (Oct 2017 – Jun 2019)
- Korean (Native), English (Native), Chinese (Native), Hebrew (Beginner)
- Multicultural background with experience living in Korea, China (11 years), and Israel (6.5 years)